



DIGITAL LITERACY

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COURSE OUTLINE

- Introduction to ICT
- History of Computers
- Components of a Computer
- Data Security
- Computer Crime
- Laws governing protection of Data in Kenya
- Applying computer software in simplifying tasks
- Communicating using internet & email
- Desktop publishing
- Presentations

DIGITAL - LITERACY

1. WHAT IS LITERACY?

Dictionaries define literacy as the ability to read and write. Within education, literacy is understood as the ability to read, write, and use arithmetic; the emphasis is on proficiency with language and numeracy.

2. WHAT IS DIGITAL ?

In this context, it refers to something (such as a device) characterized or operated by digital technology(device that can generate, process or store data)

The technology that you use and may even take for granted is not the same technology that your grandparents use or that students in another country use



DIGITAL LITERACY

Digital literacy is the skill of being able to use computers and the internet to find, evaluate, use, share, and create material.

We need to know how to use technology well in order to reach our goals and go about our daily lives, especially now that we live in a connected, online world.

PROCESS OF DEVELOPING DIGITAL LITERACY





TYPES OF DIGITAL LITERACY

1. Computer Skills

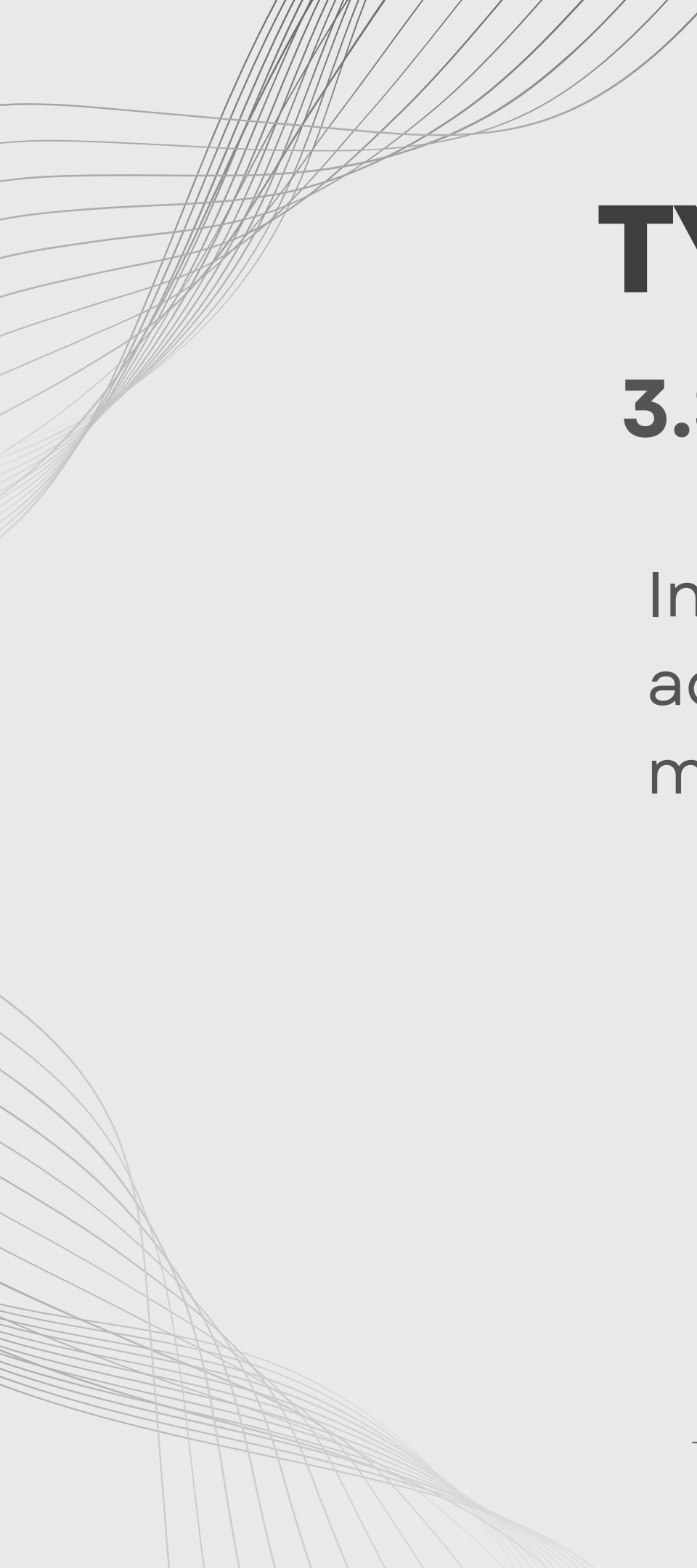
The understanding of a computer's hardware and software, as well as its information processing and delivery processes, is frequently disregarded by individuals who do not engage in a thorough exploration of this topic



TYPES OF DIGITAL LITERACY

2. Communication Skills

Gaining this talent is of utmost importance and the primary motivation for many persons' desires to go online. Utilizing platforms like Gmail to create an email account or establishing an internet video conferencing profile through websites like Skype can greatly improve your ability to engage and establish connections with others.



TYPES OF DIGITAL LITERACY

3.Social Media Skills

In order to enhance one's online social presence, it is advisable to establish accounts on the popular social media platforms like TikTok, X, and Instagram.



TYPES OF DIGITAL LITERACY

4. Information Literacy

The acquisition of proficiency in conducting through searches and employing critical analysis techniques to assess the credibility and pertinence of digital material.



TYPES OF DIGITAL LITERACY

5. Cyber Security Literacy

Gaining knowledge on the subject of online safety, comprehending the significance of privacy, and embracing conscientious actions to safeguard personal information and traverse the realm of digital hazards.



HISTORY OF COMPUTER

Watch this Video :

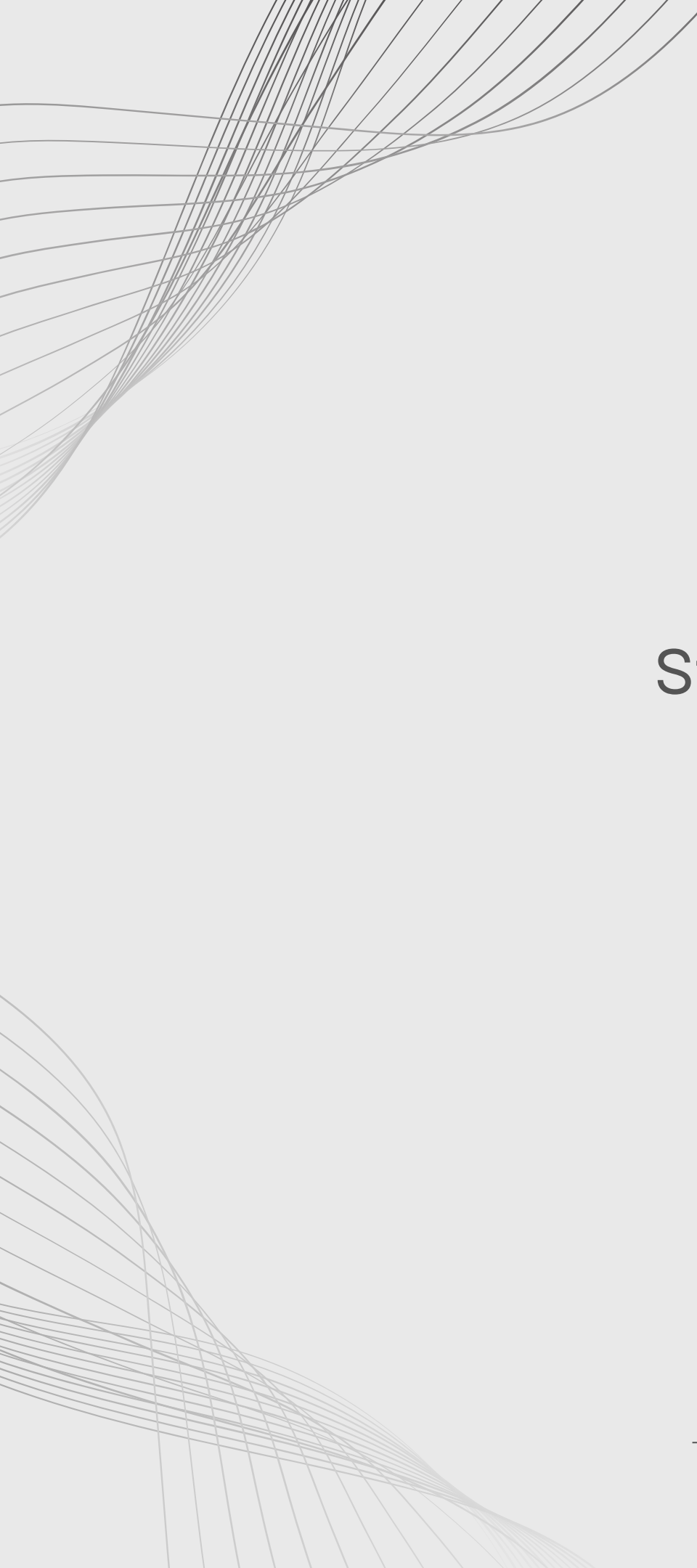
**History Of Computer | Full History And Evolution Of
Computers Till Date by Invention Flex**

<https://www.youtube.com/watch?v=Ow1BLT29p9w>



REFERENCES

- Pressbooks; Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis vel dolor ante. Nullam feugiat egestas elit et vehicula.
- ManonmaniamSundaranarUniversity(2023), ManonmaniamSundaranarUniversity.
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ASSIGNMENT

State & Explain the technologies used in each generation (1st - 5th Generation)

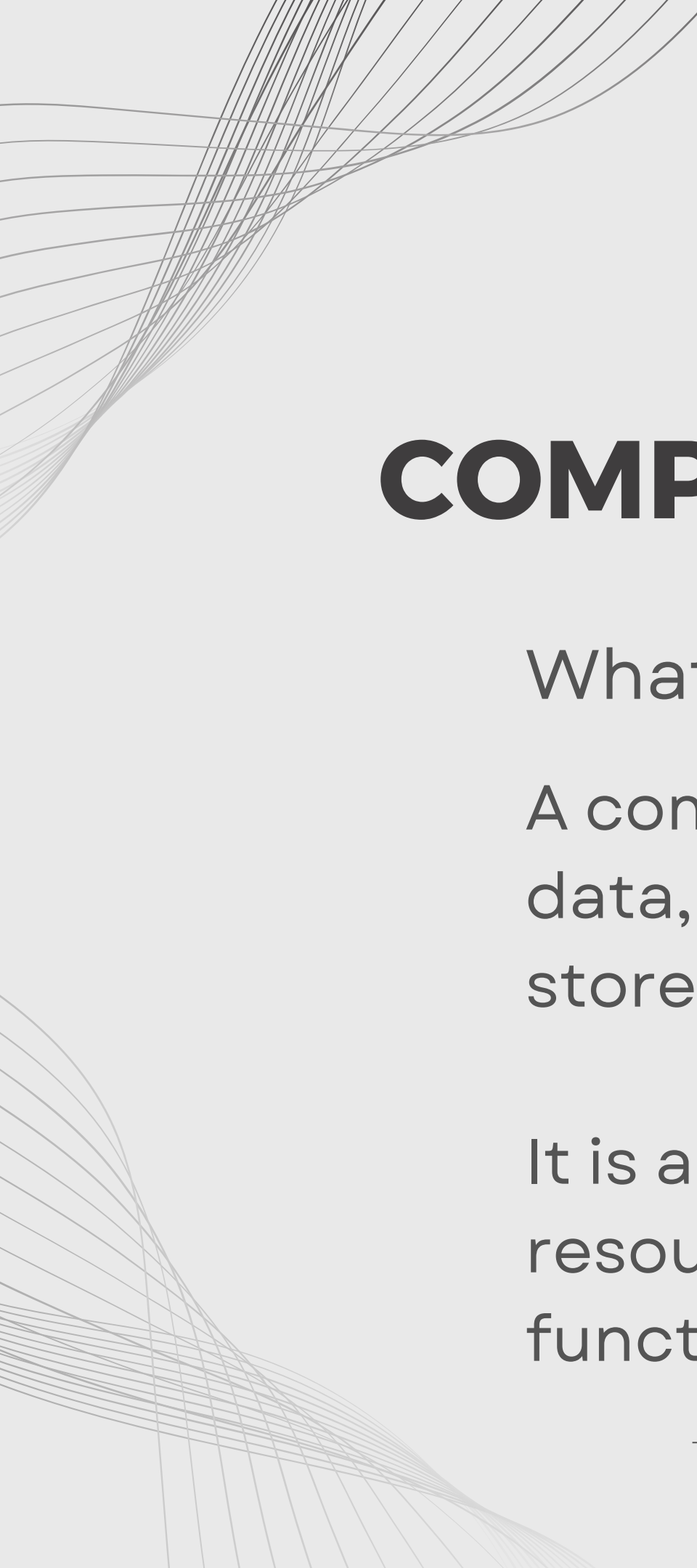


THANK YOU



RECAP

1. What is digital literacy?
2. State the types of digital literacy
3. Over time the computer has evolved across how many generations?



COMPONENTS OF A COMPUTER

What is a computer ?

A computer is an electronic device that accepts data, performs operations, displays results, and stores the data or results as needed.

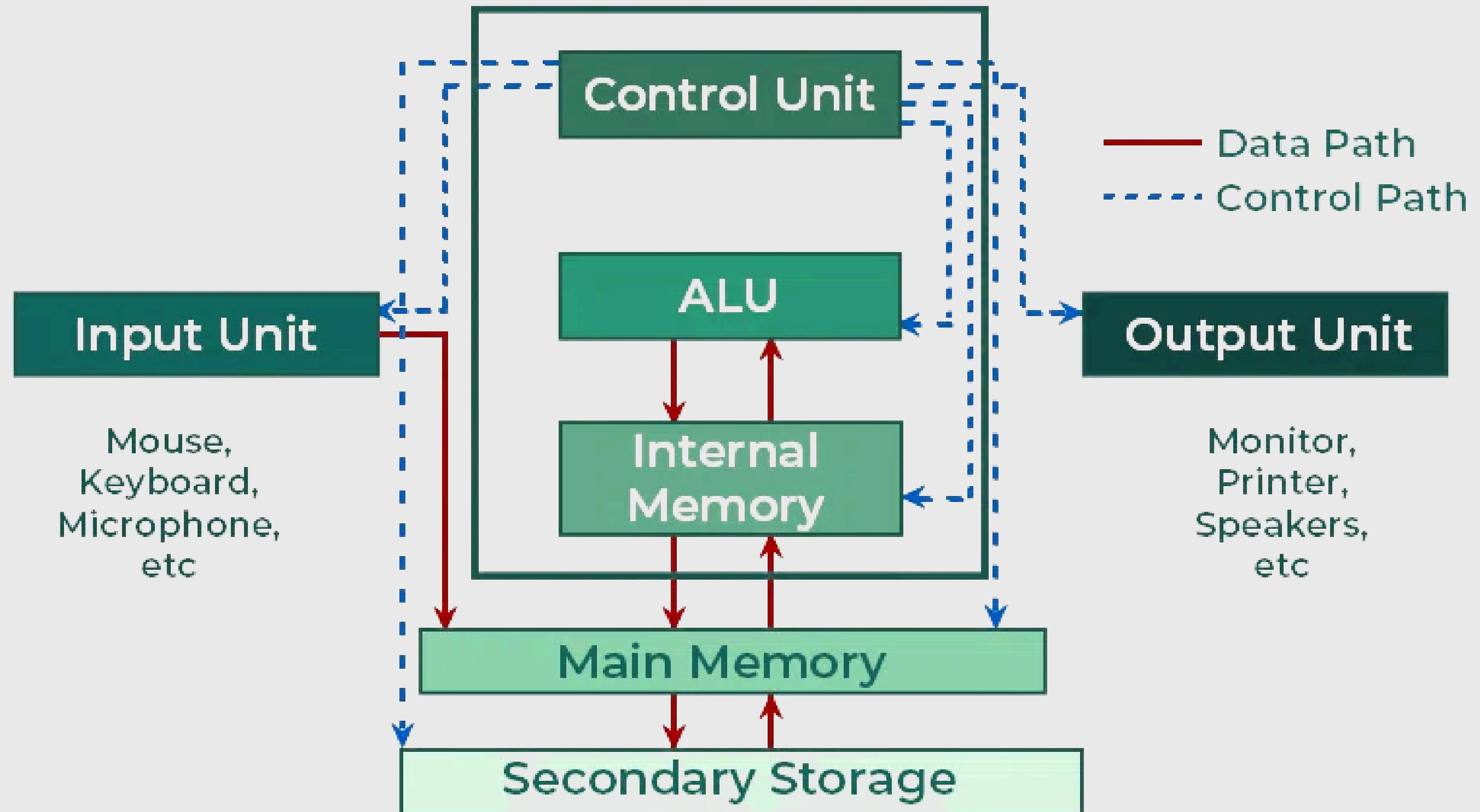
It is a combination of **hardware and software** resources that integrate and provide various functionalities to the user.



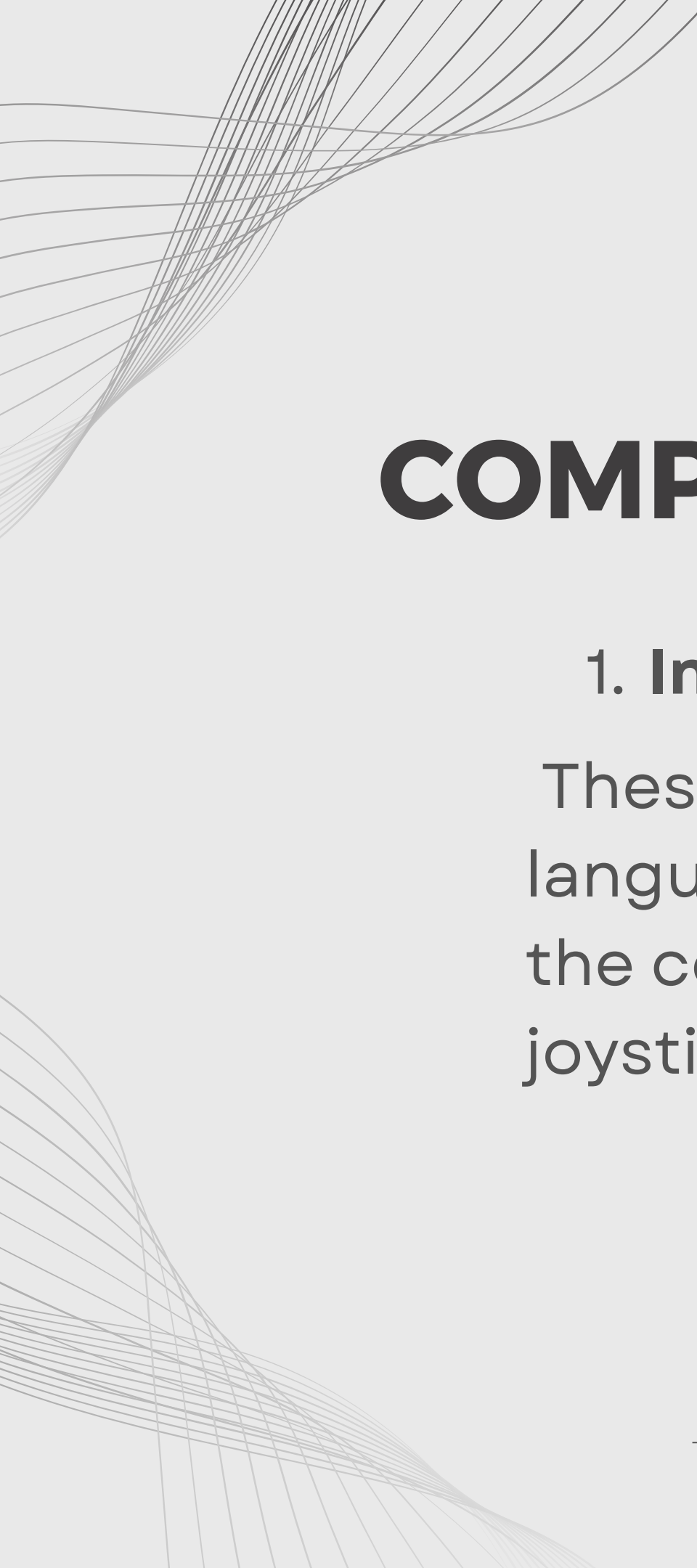
COMPONENTS OF A COMPUTER

Hardware is the physical components of a computer, such as a processor, memory devices, monitor, keyboard, etc., while **software** is a set of programs or instructions that are required by the hardware resources to function properly

Components of computer



Uploaded from [geeksforgeeks](https://www.geeksforgeeks.org/)



COMPONENTS OF A COMPUTER

1. Input Unit

These devices take input and convert it into binary language that the computer understands. Some of the common input devices are keyboard, mouse, joystick, scanner etc.

2. Central Processing Unit

This is the brain of the computer because it controls everything the computer does. When you enter information using an input device, the CPU processes it.

The CPU has three main parts:

- **Arithmetic Logic Unit** - It performs mathematical calculations and takes logical decisions.
- **Memory Register** - A register is a small, temporary memory inside the CPU. The processor uses it to store data that it is currently working on.
- **Control Unit** - It coordinates and controls the data flow in and out of the CPU, and also controls all the operations of ALU, memory registers and also input/output units



COMPONENTS OF A COMPUTER

3. Output Unit

The output unit consists of output devices that are attached to the computer. It converts the binary data coming from the CPU to human understandable form. The common output devices are monitor, printer, plotter, etc.



COMPONENTS OF A COMPUTER

4. Motherboard

The motherboard is like the backbone of a computer, connecting all the important parts such as the CPU, memory, and storage. It also helps distribute power, transfer information, and connect devices like a mouse, keyboard, or monitor. If there is a problem with the motherboard, a computer technician may open the PC to check for any loose or damaged connections, such as corrosion.



COMPONENTS OF A COMPUTER

5. Random Access Memory

RAM is the computer's short-term memory, where data is stored temporarily while programs are running. For example, when you open an app, it loads into RAM so the computer can access it quickly.



COMPONENTS OF A COMPUTER

6. Storage Device

A storage device provides the memory for installing programs and saving files. Unlike RAM, a storage device's content is secured in non-volatile memory, meaning that data is saved permanently inside its memory bank, preserving them even after you turn off the PC

PCs use 2 common types of internal storage devices: hard disk drives (HDDs) and solid-state drives (SSDs). Other secondary storage devices exist, such as USB flash drives and portable HDDs.



COMPONENTS OF A COMPUTER

PRACTICALS

Identify the different parts of computer and state the function of each component

REFERENCES

- HP, 5 Most Important Components of a Computer;
<https://www.hp.com/my-en/shop/tech-takes/post/most-important-computer-components>
- Geek for geeks, Components of a computer;
<https://www.geeksforgeeks.org/computer-organization-architecture/computer-and-its-components/>



RECAP

- State the components of a computer
- What is the difference between output & input units
- Explain the functions of a CPU
- What is the difference between SSD HDD



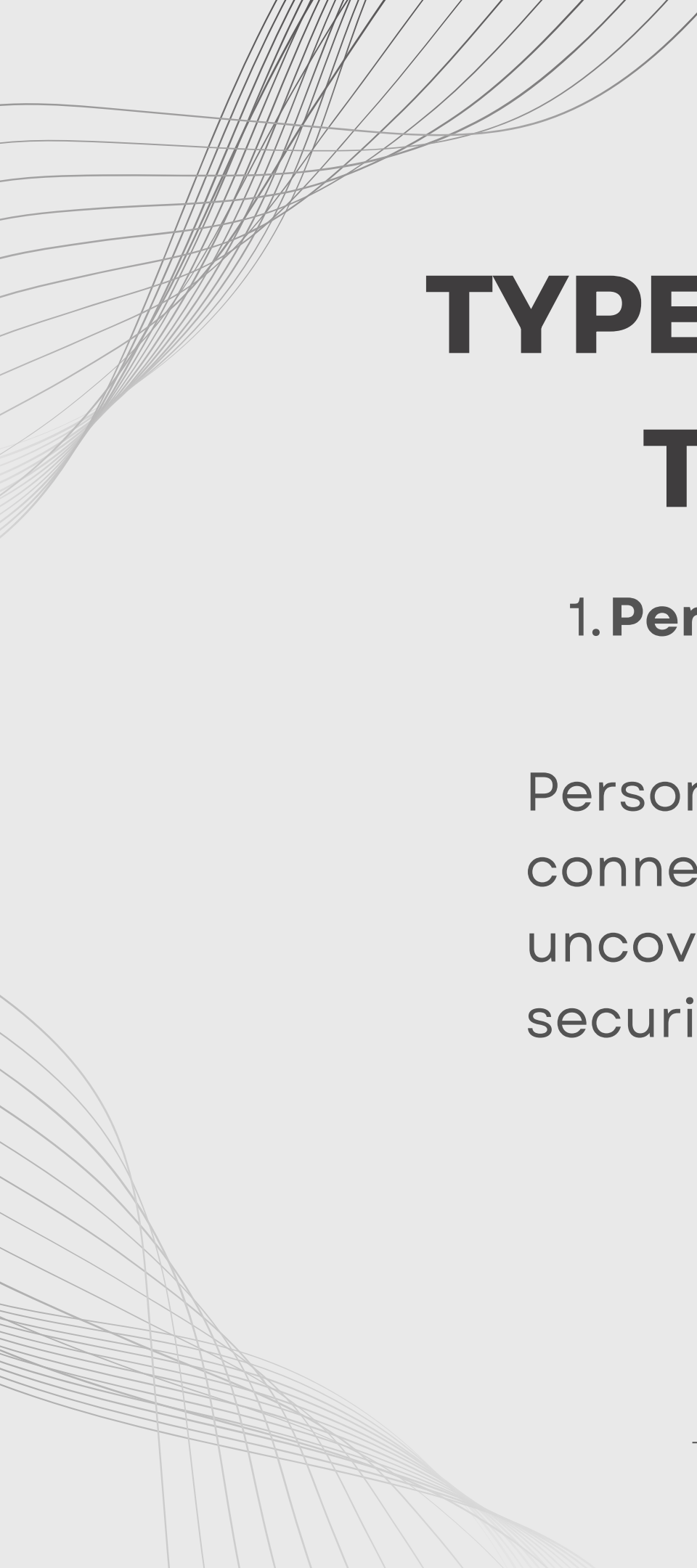
DATA SECURITY

Data Security is about protecting digital information from unauthorized access, corruption, or theft. In healthcare, it involves protecting patient records (Electronic Health Records – EHRs) and ensuring compliance with laws and ethics



IMPORTANCE OF DATA SECURITY IN HEALTHCARE

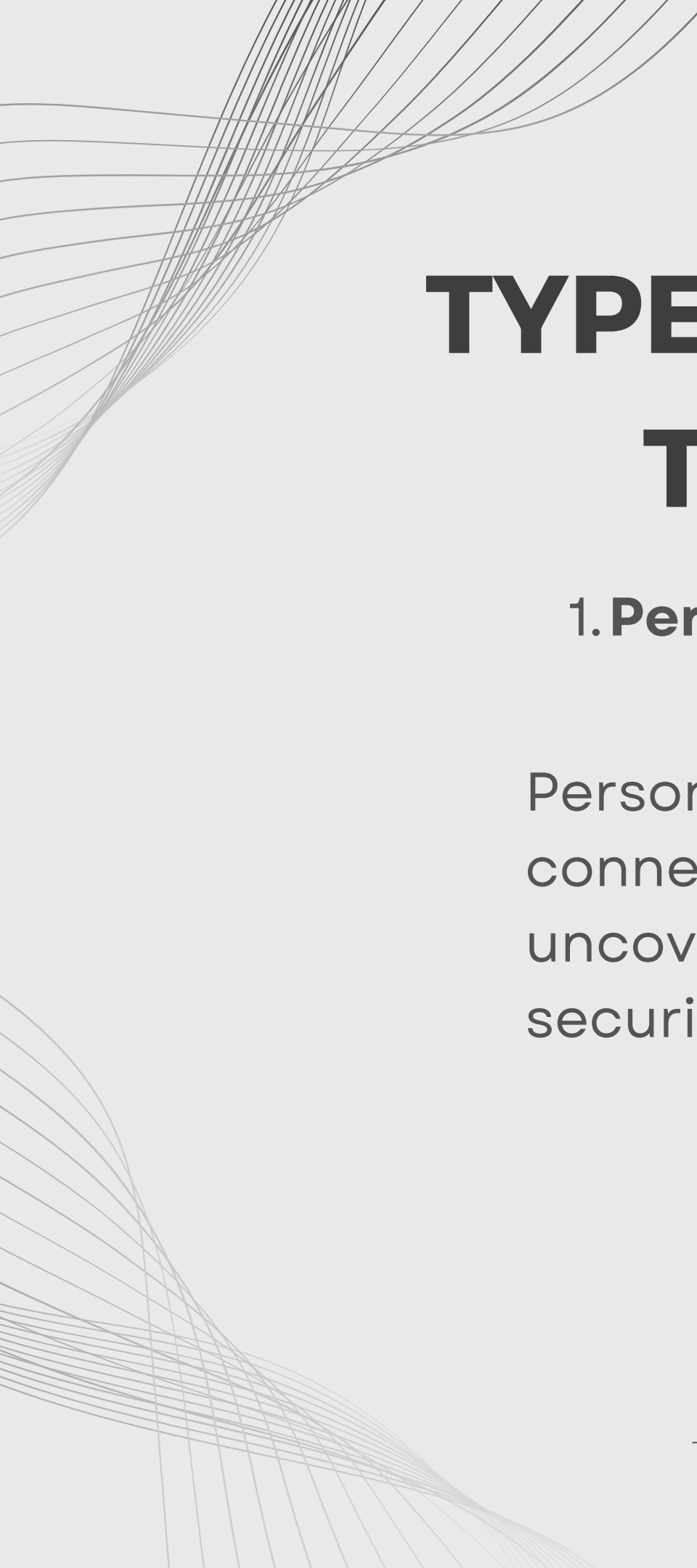
- Protects Patient Privacy
- Ensures compliance with laws (HIPAA, GDPR, Data Protection Act)
- Maintains patient trust
- Prevents financial loss
- Ensures service continuity against disruptions like ransomware



TYPES OF DATA IN HEALTHCARE THAT NEED PROTECTION

1. Personal Identifiable Information (PII)

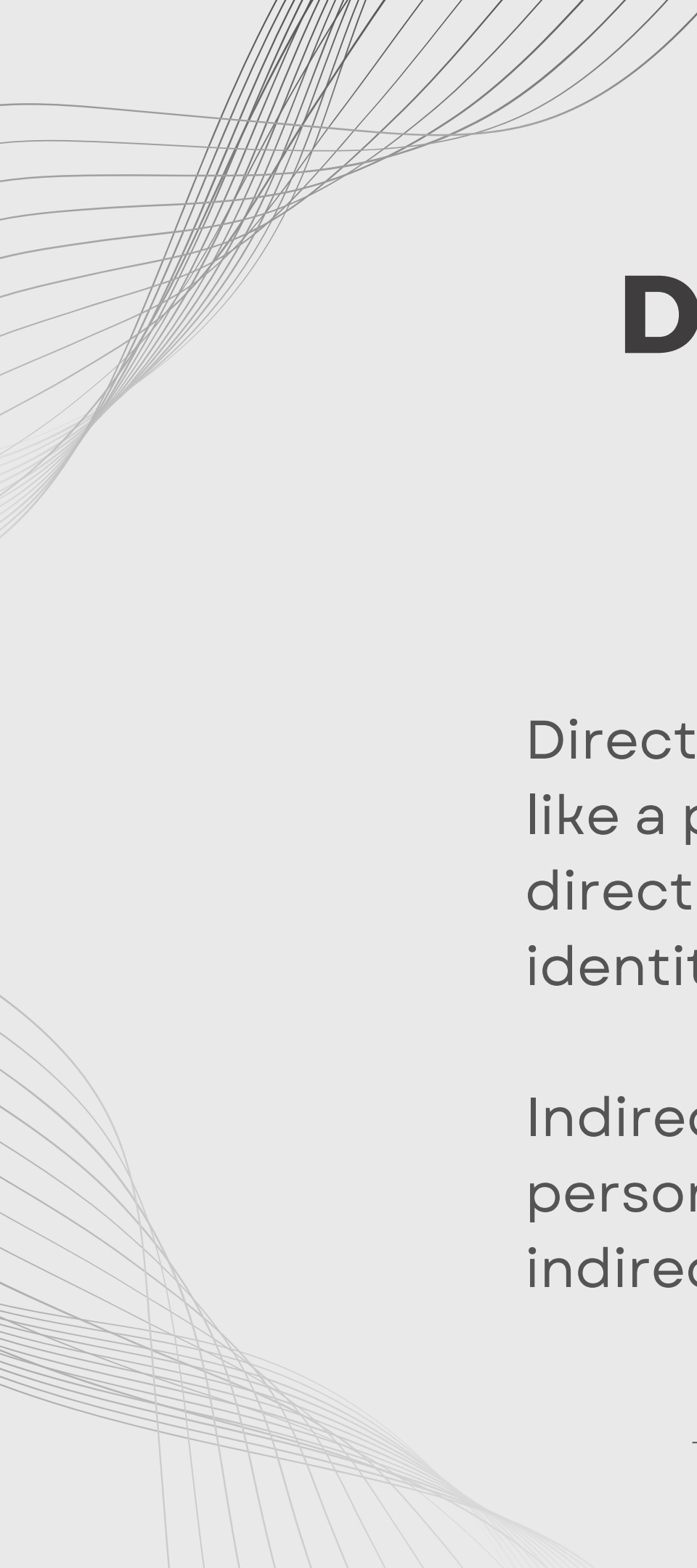
Personally identifiable information (PII) is any information connected to a specific individual that can be used to uncover or steal that individual's identity, such as their social security number, full name, email address or phone number.



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DIRECT VERSUS INDIRECT IDENTIFIERS

Direct identifiers are unique to a person and include things like a passport number or driver's license number. A single direct identifier is typically enough to determine someone's identity.

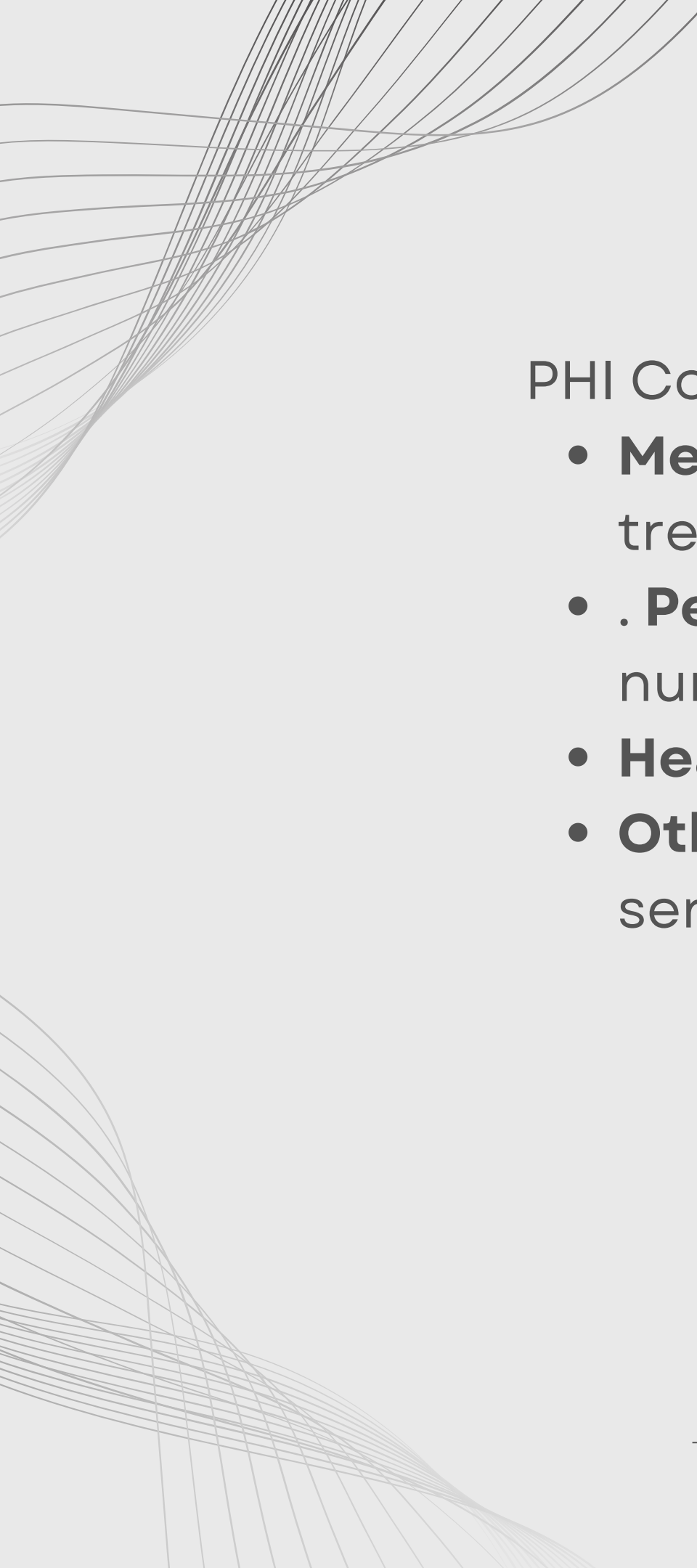
Indirect identifiers are not unique. They include more general personal details like race and place of birth. While a single indirect identifier can't identify a person, a combination can



TYPES OF DATA IN HEALTHCARE THAT NEED PROTECTION

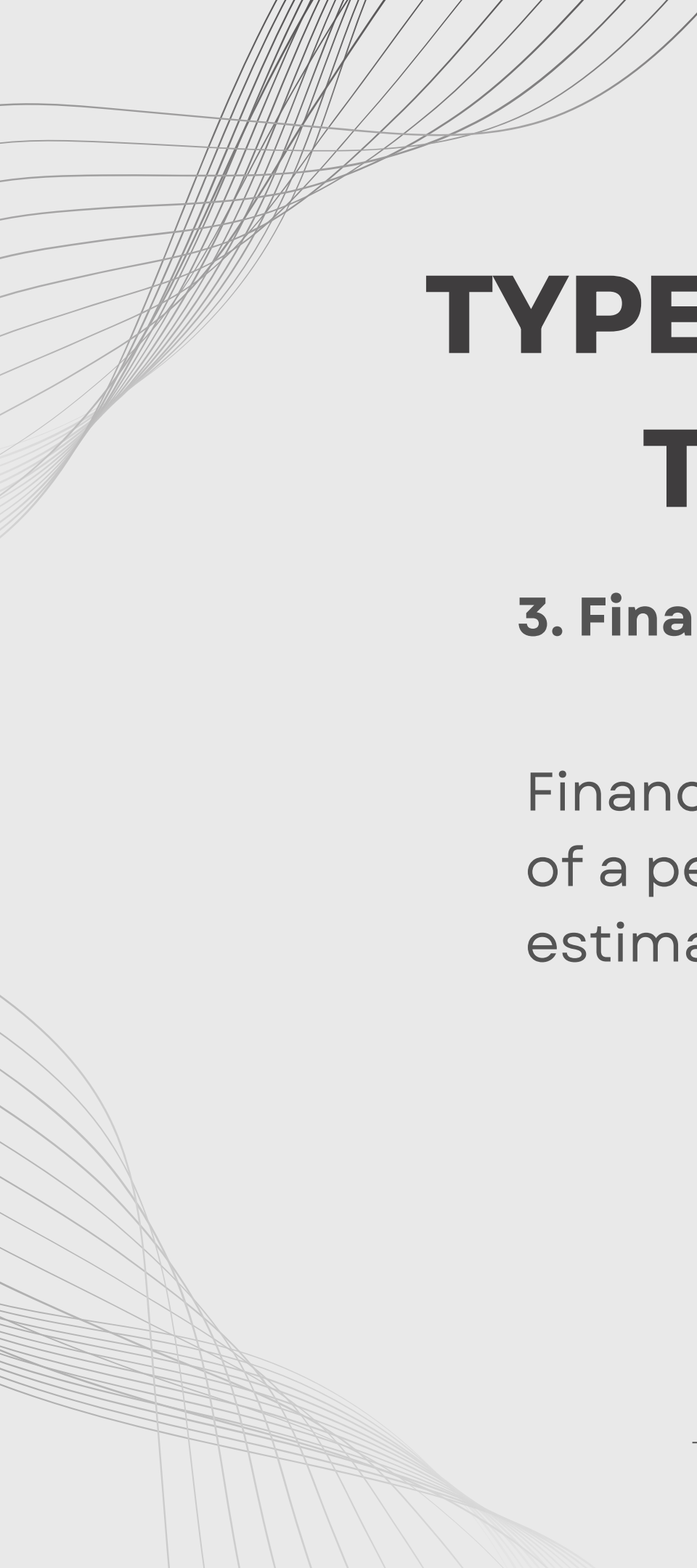
2. Protected Health Information

Protected Health Information (PHI) is any health-related information that can be used to identify an individual, including medical history, diagnoses, treatments, and identifiers like Social Security or medical record numbers.



PHI Consists of:

- **Medical Information:**Past medical history, diagnoses, treatments, test results, and prescribed medications
- **Personal Identifiers:**Names, addresses, Social Security numbers, email addresses, and phone numbers.
- **Health Plan Information:**Health plan beneficiary numbers.
- **Other Health-Related Data:**Information about health services, and payments for those services.



TYPES OF DATA IN HEALTHCARE THAT NEED PROTECTION

3. Financial Information

Financial information is data about the monetary transactions of a person or business. This information is use to derive estimates of credit risk by creditors and lenders.

PRINCIPLE OF DATA SECURITY

CIA

What Does CIA Stand For?



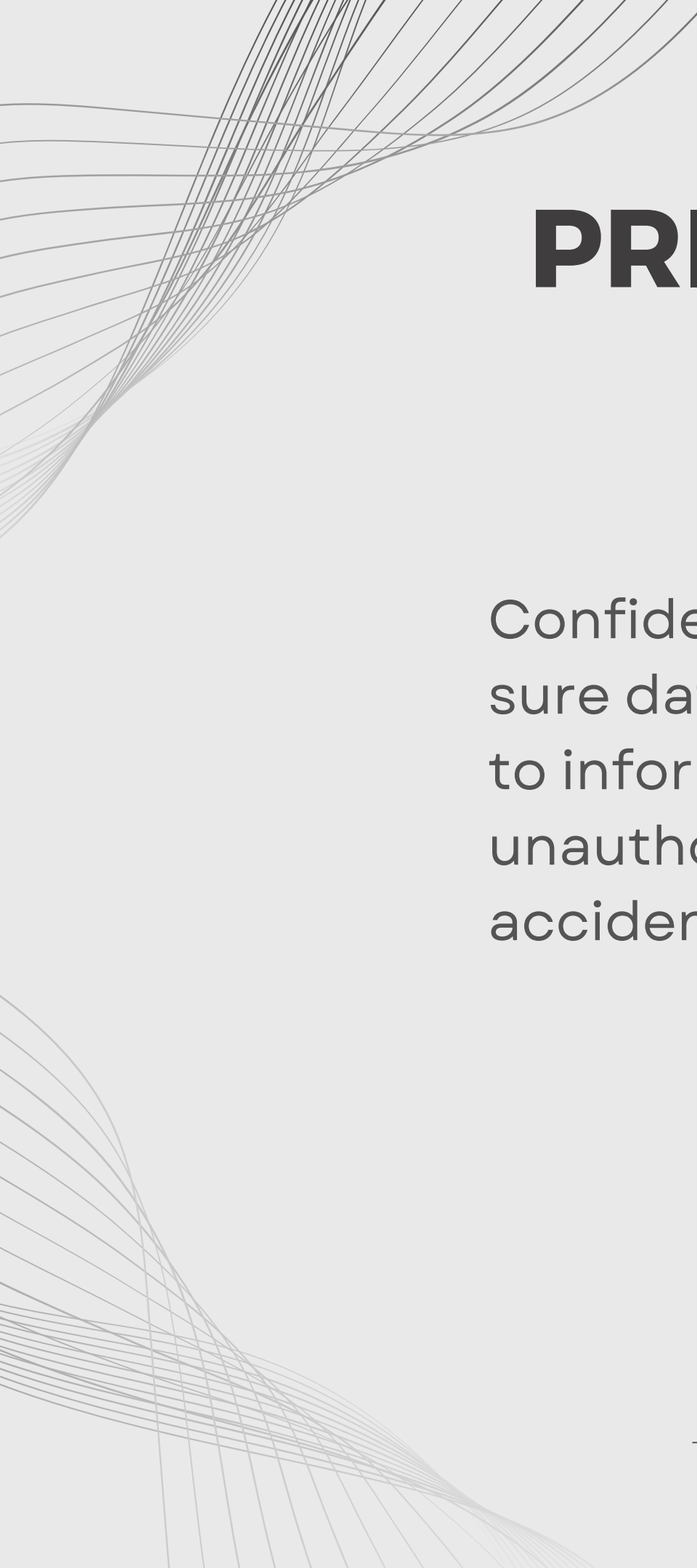
Confidentiality



Integrity



Availability



PRINCIPLE OF DATA SECURITY

CONFIDENTIALITY

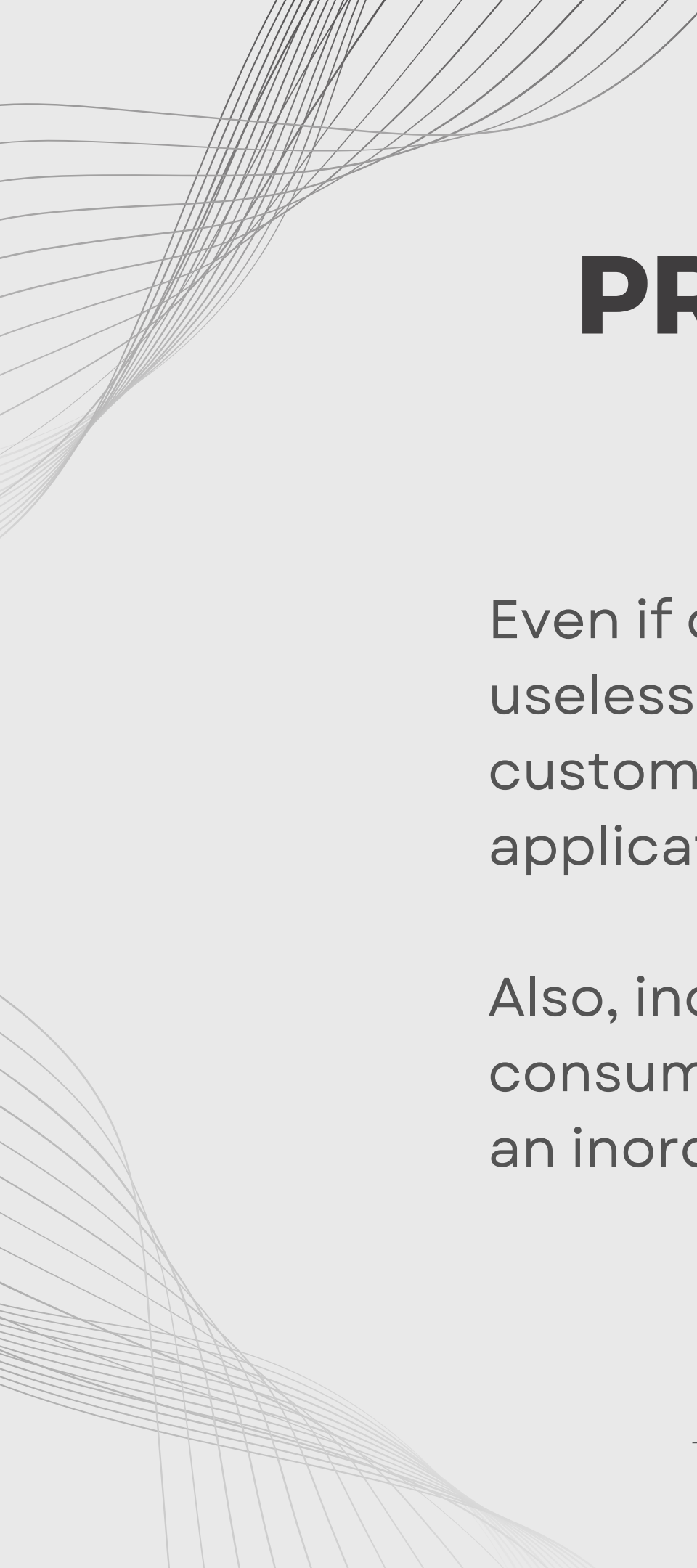
Confidentiality involves the efforts of an organization to make sure data is kept secret or private. To accomplish this, access to information must be controlled to prevent the unauthorized sharing of data—whether intentional or accidental.



PRINCIPLE OF DATA SECURITY

INTEGRITY

Integrity involves making sure your data is trustworthy and free from tampering. The integrity of your data is maintained only if the data is authentic, accurate, and reliable.



PRINCIPLE OF DATA SECURITY

AVAILABILITY

Even if data is kept confidential and its integrity maintained, it is often useless unless it is available to those in the organization and the customers they serve. This means that systems, networks, and applications must be functioning as they should and when they should.

Also, individuals with access to specific information must be able to consume it when they need to, and getting to the data should not take an inordinate amount of time.



THREATS TO DATA SECURITY IN HEALTHCARE

- Internal Threats – Unauthorized staff access, mishandling
- External Threats – Hackers, ransomware, phishing
- Physical Threats – Stolen devices, natural disasters